

Impact of NEWS2 severity and early management decisions on community-onset sepsis outcomes: an observational cohort study

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Impact and Importance

Sepsis affects 246,000 people and causes 50,000 deaths in the UK¹.

Rapid, effective antibiotics can reduce mortality by 3x-4x.

Current broad-spectrum guidelines lack robust evidence for optimising initial choice based on individual risk and the heterogeneity of infection source².

Evaluating the interplay between infection source, prescribing and physiological severity (NEWS2) is essential to refining clinical management and antimicrobial stewardship.

Aims

This study aims to characterise UK empiric prescribing patterns and quantify the association between adequate therapy and 30-day mortality, while determining the impact of infection source and assessing NEWS2 bands as a prognostic tool for mortality in a real-world sepsis cohort.

Methodology

Population was derived based on NICE guidance³ for suspected sepsis from an electronic health record dataset in a tertiary hospital network (January 2021 to March 2025).

Cohorts were defined by suspected infection source (non-urinary vs. urinary) and risk-stratified by worst NEWS2 score upon admission.

Multivariable logistic regression was performed to identify the independent effect of inadequate empiric therapy on 30-day mortality.

Results

- Total of 16,151 adult patients meeting NICE criteria for suspected community-onset sepsis
- 84.1% (n=13,586) remained culture-negative and 15.9% (n=2,565) were microbiologically confirmed
- Overall community-onset sepsis confirmed rate of 0.60 per 1,000 bed days

1. Frequent transitions between antimicrobial UK-AWaRe categories occur during first 7 days of admission

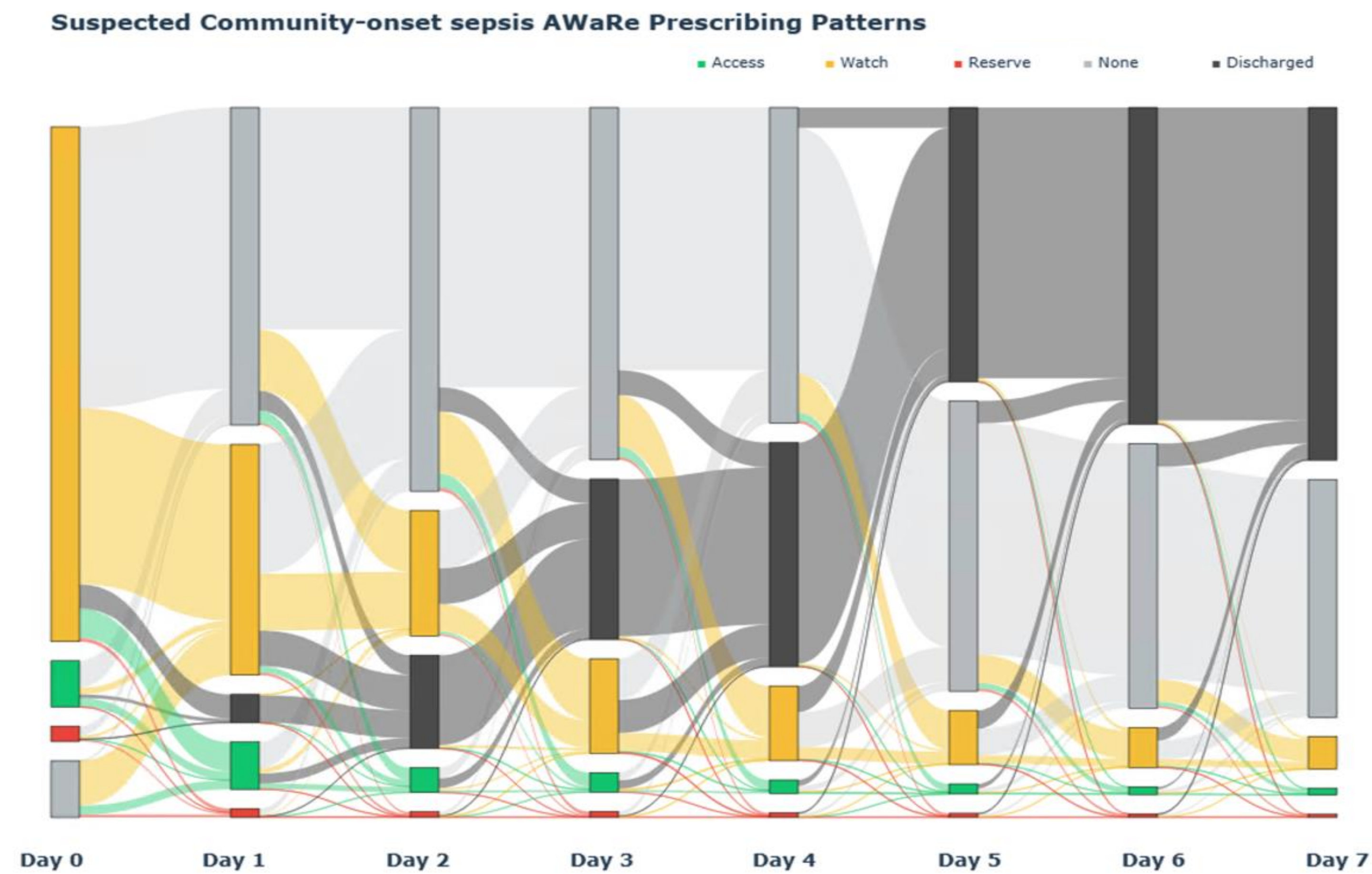


Figure 1. Daily transitions of UK-adapted AWARe categories from admission (Day 0) to Day 7

2. Each unit increase in admission NEWS2 score significantly increased odds of 30-day mortality (p<0.001) for non-urinary source (OR 1.22; 95% CI 1.21–1.24) and urinary source sepsis Cohort (OR 1.19; 95% CI 1.15-1.24)

NEWS2 Category	Non-urinary source sepsis (No. of distinct admissions = 22,606)		Urinary Source Sepsis (No. of distinct admissions = 3,397)	
	Mortality	NEWS2 Comparison	Mortality	NEWS2 Comparison
Overall	11.2% (2,532)	—	9.2% (312)	—
≥ 3	12.5% (18,456)	< 3: 5.3% (4,150)	10.3% (2,737)	< 3: 4.5% (660)
≥ 5	15.8% (11,630)	< 5: 6.3% (10,976)	13.4% (1,640)	< 5: 5.3% (1,757)
≥ 7	19.4% (6,728)	< 7: 7.7% (15,878)	15.9% (897)	< 7: 6.8% (2,500)
NEWS2 Bands (NICE Guidance)				
0 (Low)	2.2% (758)	—	2.0% (102)	—
1–4 (Low)	6.6% (10,218)	—	5.5% (1,655)	—
5–6 (Moderate)	11.0% (4,902)	—	10.4% (743)	—
≥ 7 (High)	19.4% (6,728)	—	15.9% (897)	—

Data are presented as % (n).

3. Impact of empiric adequate therapy on survival is infection source dependent: essential in non-urinary, yet not significant urinary source

Variable	Non-urinary source (n = 2291)			Urinary-Source (n = 849)		
	OR	P-Value	Significance	OR	P-Value	Significance
Age	1.05	$p < 0.0001$	Significant	1.05	$p < 0.0001$	Significant
NEWS2 Score	1.20	$p < 0.0001$	Significant	1.20	$p < 0.0001$	Significant
Malignancy	1.49	$p < 0.05$	Significant	2.08	$p < 0.05$	Significant
Empiric Adequacy	0.71	$p < 0.005$	Significant	1.49	0.5860	Not Significant
Diabetes	0.93	0.6000	Not Significant	0.74	0.3886	Not Significant
CKD	0.87	0.4428	Not Significant	0.43	$p < 0.05$	Significant
Frailty	1.17	0.5493	Not Significant	1.61	0.3862	Not Significant
Dementia	1.41	0.0917	Not Significant	1.87	0.2096	Not Significant
Cardiac failure	0.89	0.6847	Not Significant	1.87	0.3698	Not Significant
ESBL status	0.70	0.0674	Not Significant	0.18	0.0941	Not Significant
Ethnicity	1.79	0.1316	Not Significant	1.25	0.8623	Not Significant
Gender	0.92	0.4503	Not Significant	1.09	0.7422	Not Significant
IMD	0.56	0.1841	Not Significant	1.21	0.8744	Not Significant

Note: OR = Adjusted Odds Ratio. NEWS2 = National Early Warning Score 2. CKD = chronic kidney disease

4. Inadequate therapy was associated with increased odds of death (aOR 1:43; 95% CI 1.14-1.81; p<0.005), primarily driven by non-ESBLs (n=2,271; aOR 1.39; 95% CI 1.10-1.74; p=0.005)

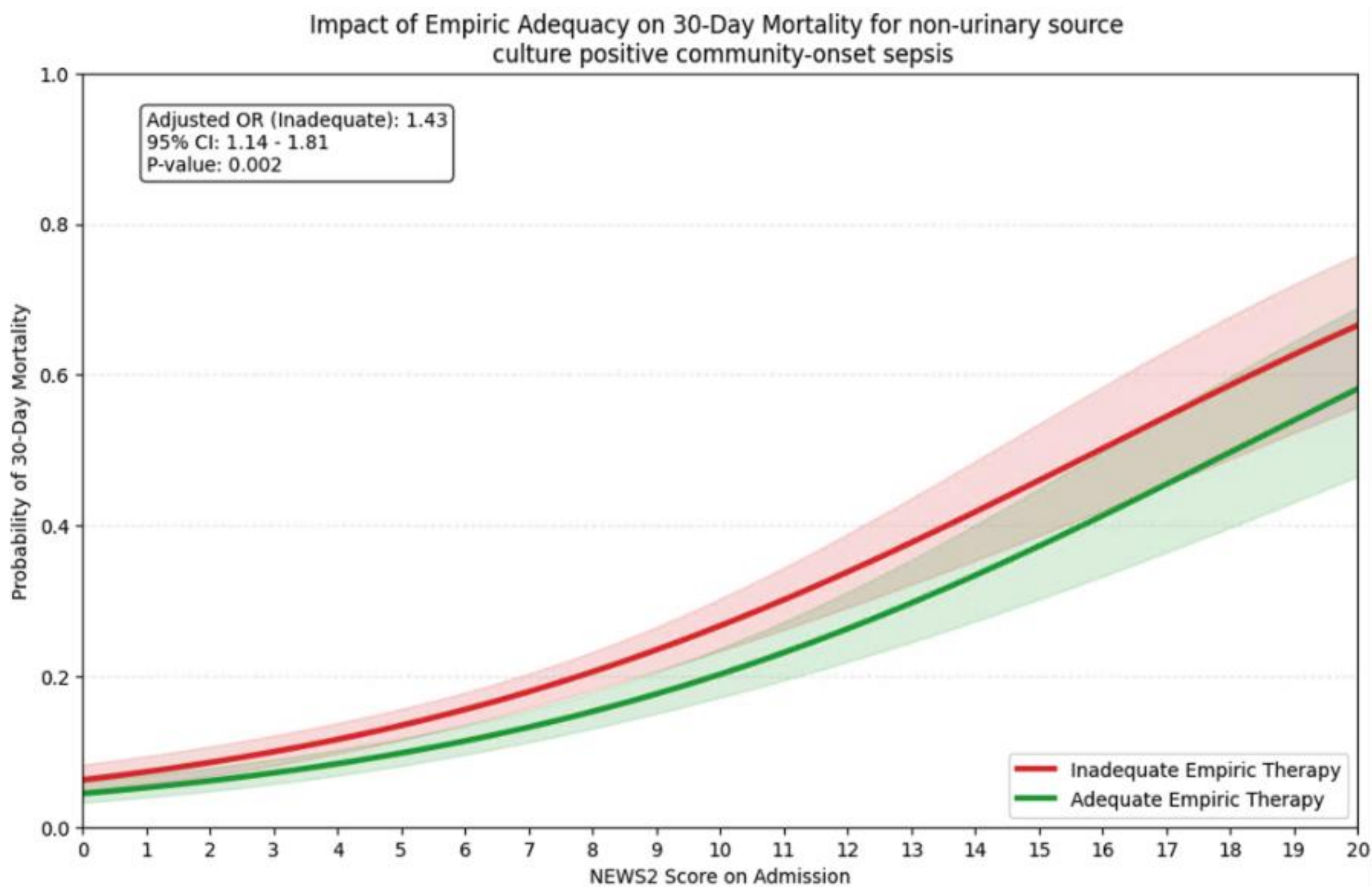


Figure 2. Impact of empiric antimicrobial adequacy on 30-day mortality for non-urinary source.

Conclusions

Validating NEWS2 across all cohorts addresses NICE research recommendation gaps, showing mortality is driven by antimicrobial adequacy in non-urinary sepsis (p=0.002) but by host factors (age and comorbidities) in urinary source. Consequently, optimising community-onset sepsis management requires enhanced risk profiling and pre-admission data integration to improve screening and prevent sepsis-related mortality.

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